[INCH-POUND]
A-A-59134
28 October 1997
SUPERSEDING
MIL-I-46865B (MI)
28 June 1991

COMMERCIAL ITEM DESCRIPTION

INSULATING COMPOUND, ELECTRICAL, EPOXY, COLLOIDAL SILICA FILLED FOR POTTING AND ENCAPSULATION

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-I-46865B (MI) for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description covers an epoxy resin electrical insulating compound used for potting and encapsulating

1.2 Classification. The epoxy resin shall be of the following types, as specified.

Type I - Low viscosity

Type II - Medium viscosity

Type III - High viscosity

2. SALIENT CHARACTERISTICS

2.1 Materials.

2.1.1 Composition and condition in container.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270.

FSC 5970

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2.1.1.1 Composition. The materials furnished under this commercial item description shall be premixed, frozen, degassed compounds formulated from the ingredients specified in table I.

TABLE I. Material composition.

Ingredients	Parts be weight		
	Type I	Type II	Type III
Epoxy resin I	50	50	50
Epoxy resin II	50	50	50
Amine curing agent	19.5	19.5	19.5
Colloidal silica	2.5	3.5	5.0

2.1.1.2 Condition in container. The compounds, as received, shall be free of gel particles, dirt, or any other contaminants which would affect the intended use.

2.2.2 Pot life. The useable pot life of the compounds shall be not less than 30 minutes.

2.2.3 Viscosity. The viscosity of the thawed materials, as received, shall be as follows.

Type I 300 ± 100 centipoises (cP)

Type II 850 ± 200 cP

Type III 7500 ± 2000 cP

2.2.4 Storage life. The premix, as furnished by the manufacturer, shall be capable of meeting the requirements specified herein after storage at a temperature of less than 40 °C in an original closed container for not less than 3 months.

2.3 Porosity. The cured compound shall exhibit no bubbles or porosity smaller than 0.031 inch diameter.

2.4 Hardness. The cured compound shall have a hardness value of not less than 40 Shore D as specified by and when tested in accordance with ASTM D 2240.

2.5 Flexibility. Bending shall cause no chipping or cracking.

2.6 Insulation resistance. The insulation resistance shall be not less than 1×10^{11} ohms.

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2.7 Workmanship. The compound shall be uniform in quality and shall be free of dark colored contaminants, lumps, foreign materials, and other defects that could adversely affect their use.

3. REGULATORY REQUIREMENTS

3.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Market Acceptability (MA). The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

4.2.1 The company producing the item must have been producing a product meeting the requirements of this CID for at least two years.

4.2.2 The company must have sold 1000 units meeting this CID in the commercial marketplace over the past two years.

4.3 Inspection requirements. Bid samples may be required for CID items when necessary to ensure product quality.

4.3.1 Examination. Each container in the sample shall be visually examined to verify the requirements of 5.1.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

6.1 This compound is intended for use in potting and encapsulation of low voltage (less than 300 volts) cable compositions of differing viscosities.

Type I - Low viscosity material for use where connector pins have been previously sealed

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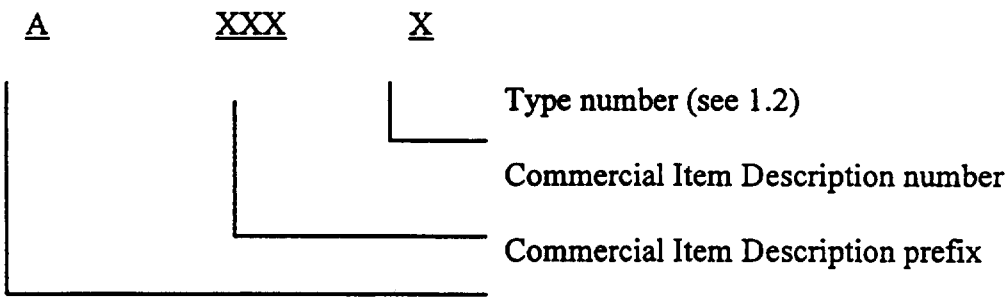
to prevent leak-through along the connector pins. Normally used when heat curing is desired.

Type II - Medium viscosity material used with or without pin sealing. If pin sealing is not performed with type II material, the type II material should be cured at room temperature.

Type III - High viscosity material for use on connectors with large open areas or widely spaced pins, where the material has poor flow characteristics.

6.2 Source of documents. The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.3 Part or identifying number.



6.4 Subject term (keyword) listing.

Hardener
Resin laminate
Premix
Crystalline substance

6.5 Product availability. This insulating compound is available commercially as EPO-TEK Epoxy Systems - 360T, 360ST, 353ND-T, and OE138 through Epoxy Technology, Inc.; and as E-Cast F-82 with 215 Hardener for Type I, E-Cast F-82 with 929 Hardener for Type II, and E-Cast Sur-Grip with 282 Hardener for Type III through the United Resin Corporation. The compound will have to be sent to a company such as APS Adhesive Packaging Specialties, Inc. to be premixed, frozen, and degassed.

Custodian:
Army - MI

Preparing Activity:
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